

Class 05: Data Viz with ggplot2

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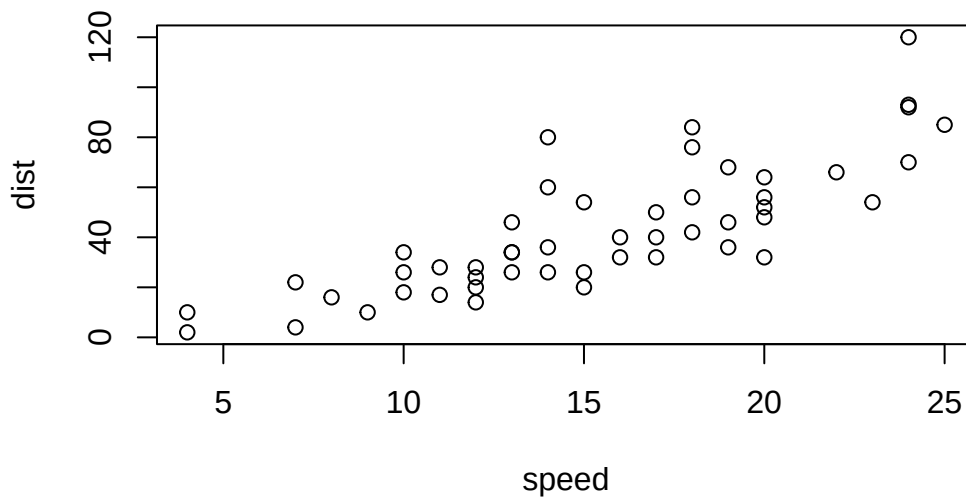
Background

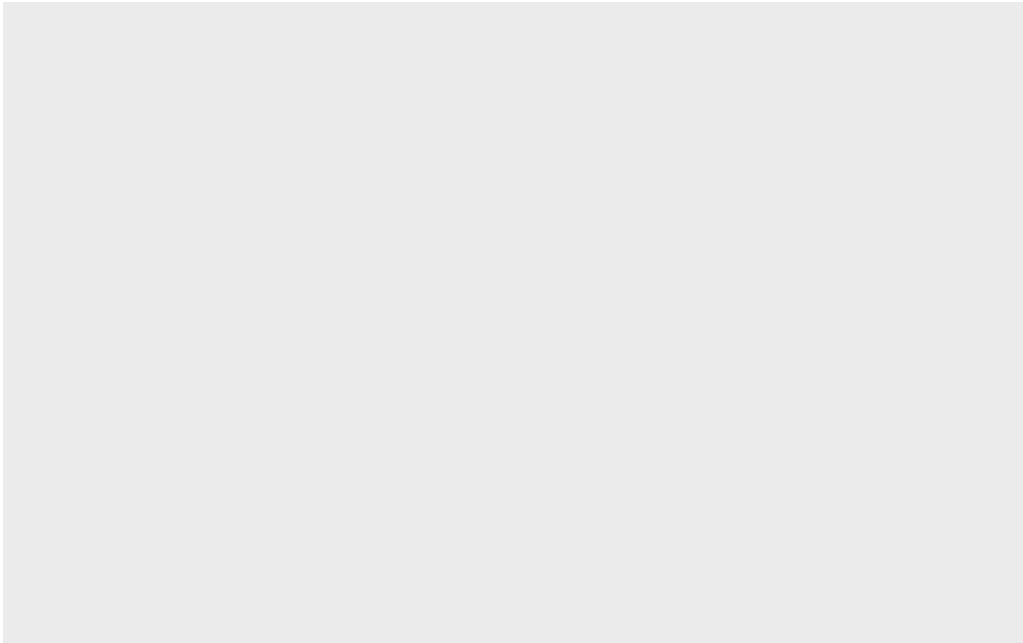
There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the ‘plot()’ function and add on packages like “ggplot2”.

Let’s compare how we make a simple figure with these two systems:

We can use the in-built ‘cars’ dataset:

```
plot(cars)
```

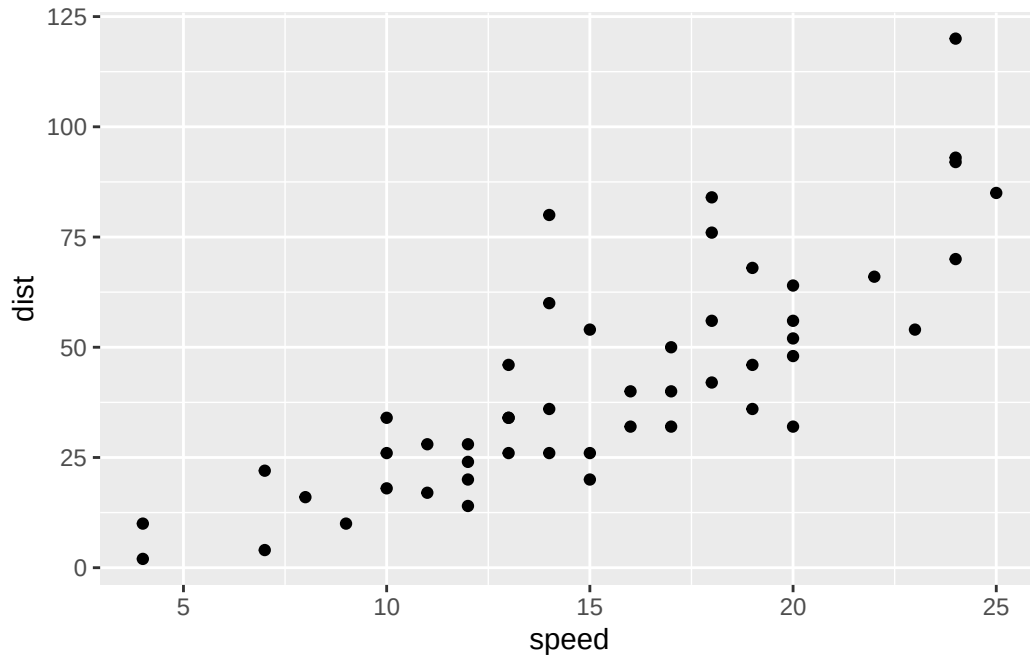




Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()
```

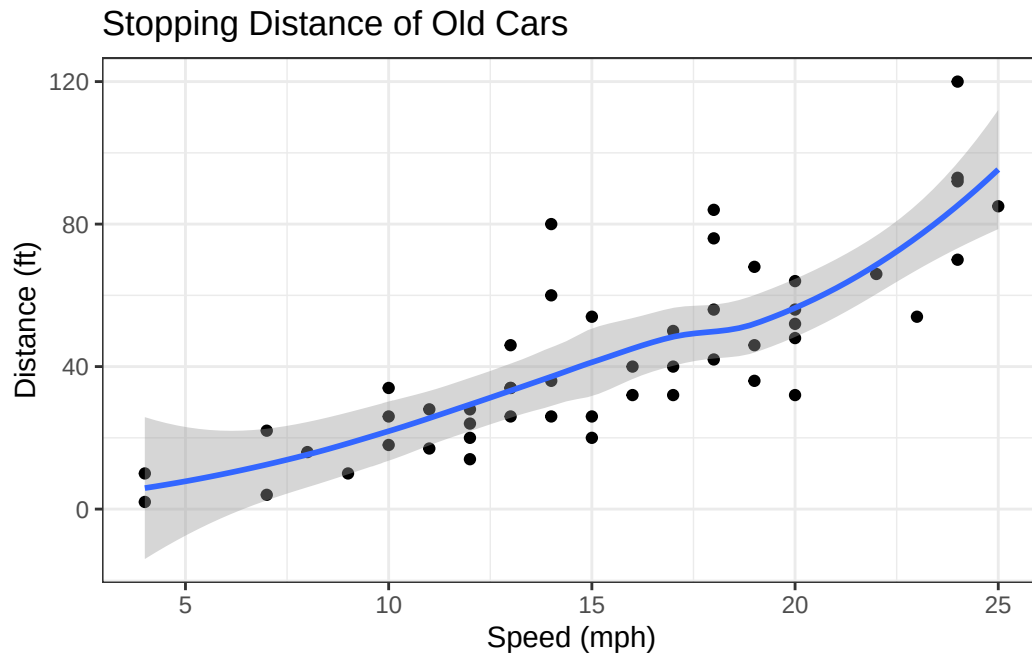


Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth() +  
  theme_bw() +  
  labs(title="Stopping Distance of Old Cars", x="Speed (mph)", y="Distance (ft)")
```

`geom_smooth()` using `method = 'loess'` and `formula = 'y ~ x'`



Q. Can you make the ‘geom_smooth()’ function produce a linear straight line fit to the data and turn off the “gray” error region.

Gene Expression Figure

Import the data plot

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

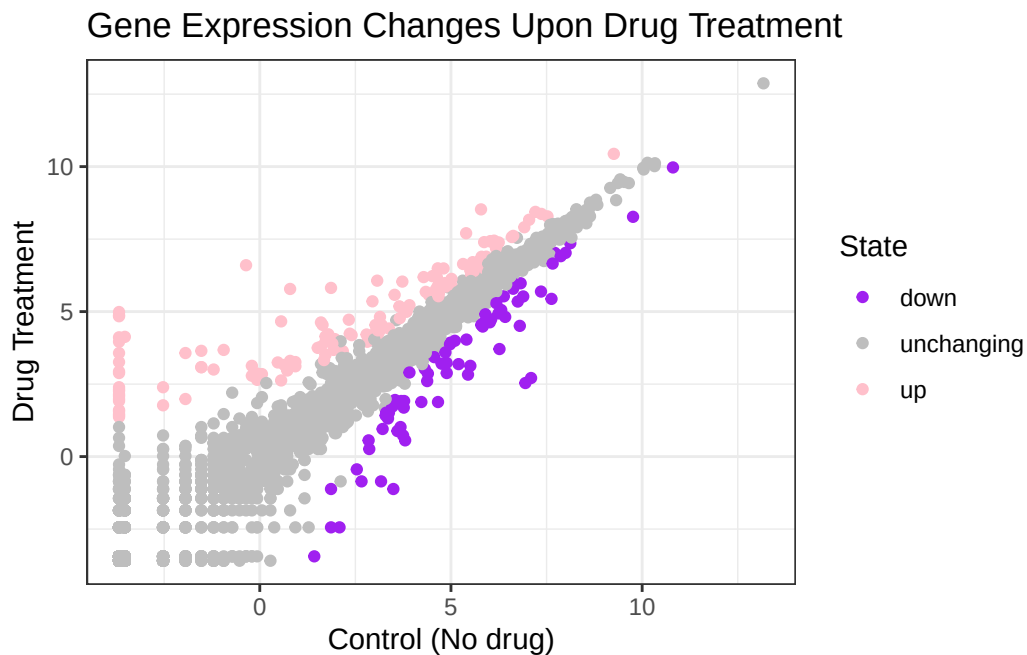
A useful new function in this context is the 'table()' function:

```
table(genes$State)
```

```
down  unchanging    up
   72      4997    127
```

My first plot attempt

```
ggplot(genes) +  
  aes(Condition1, Condition2, col=State) +  
  geom_point() +  
  scale_colour_manual(values=c("purple", "grey", "pink")) +  
  theme_bw() +  
  labs (x="Control (No drug)", y="Drug Treatment", title="Gene Expression Changes Upon Drug T
```



Going Further

Here we read the famous gapminder dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
```

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

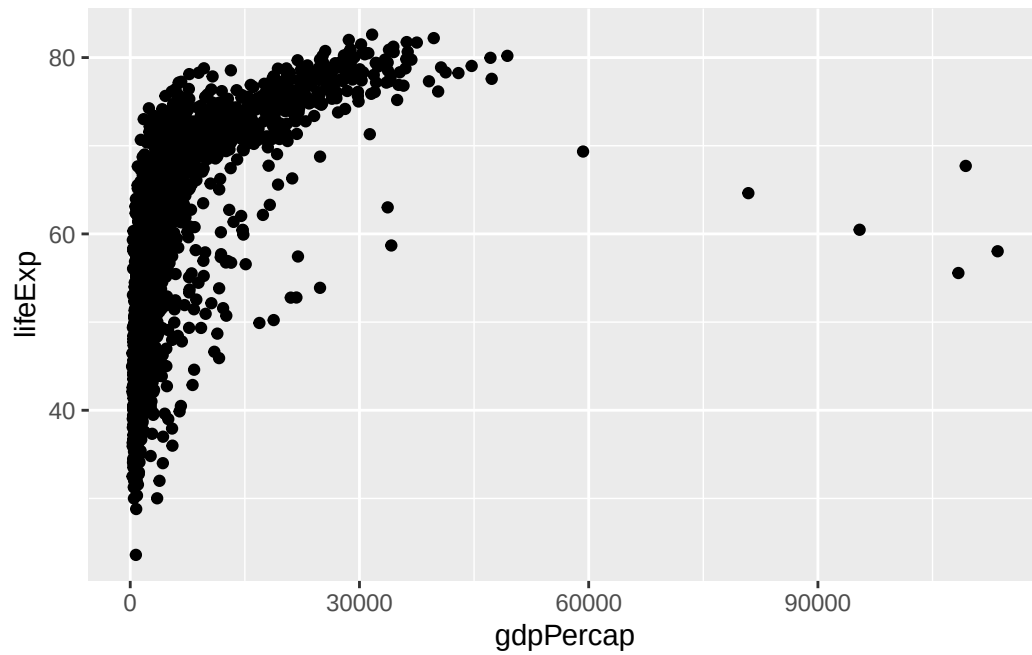
```
length(unique(gapminder$country))
```

```
[1] 142
```

Let's make our first plot of the entire dataset:

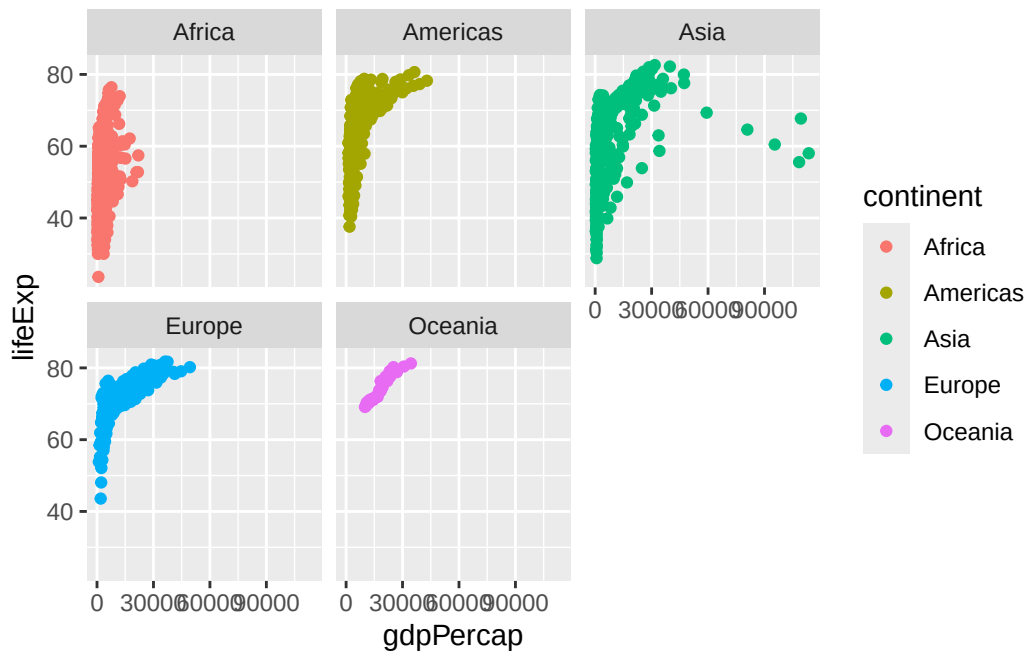
Plot of “gdpPercap” vs “lifeExp” colored by “continent”

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point()
```



I can add more layers to “p”

```
ggplot(gapminder) +  
  aes(x=gdpPercap, y=lifeExp, color=continent) +  
  geom_point() +  
  facet_wrap(~continent)
```



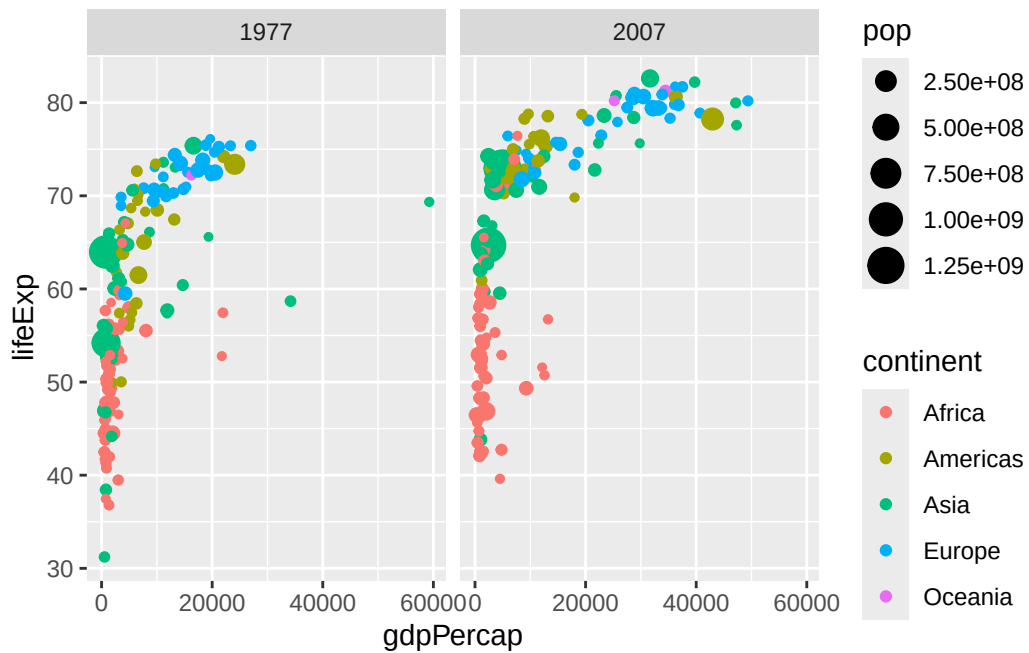
Make a plot for 1977 and 2007 only (not all the years in the dataset).

Q. First use the **dplyr** package and the 'filter()' function from that package to extract the rows from the year 2007.

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
g <- filter(gapminder, year == 2007 | year == 1977)
```

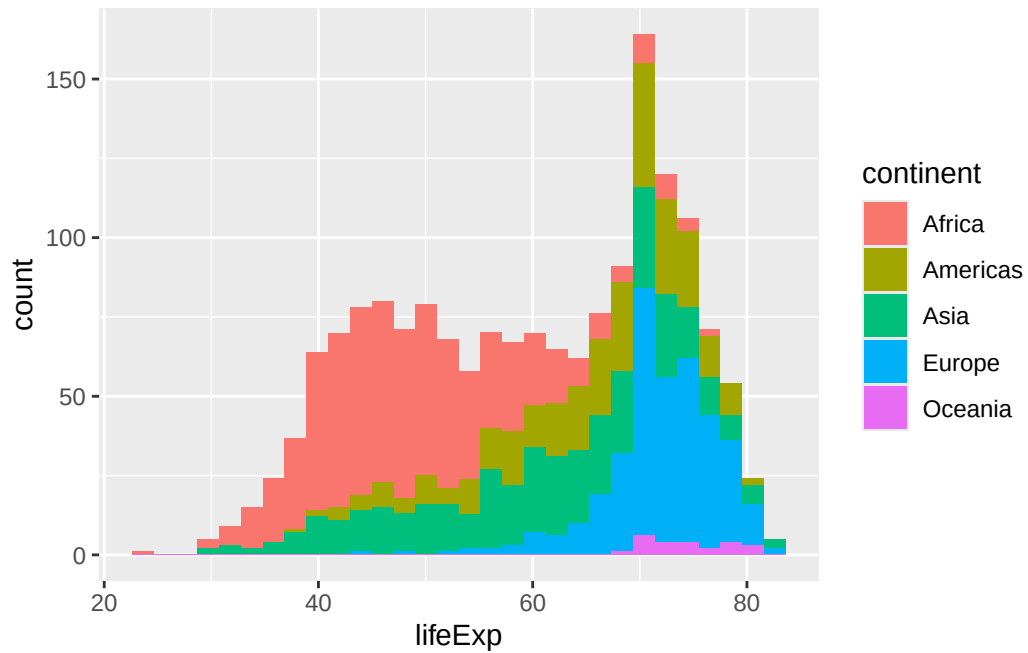
```
ggplot(g) +
  aes(x=gdpPerCap, y=lifeExp, color=continent, size=pop) +
  geom_point() +
  facet_wrap(~year)
```



Q. Make a histogram of lifeExp colored by continent

```
ggplot(gapminder)+
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder)+  
  aes(lifeExp) +  
  geom_histogram() +  
  facet_wrap("continent")
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

